

OUTILS D'AUDIT ET D'EXPLOITS RÉSEAU

Cours: Concepts et outils la sécurité des réseaux

Environnement: Kali Linux + Metasploitable (VMware)

Objectif: Se familiariser avec les outils d'audit réseau et le techniques d'exploitation

Avertissement légal

Ce TP est réalisé dans un environnement isolé et contrôlé (machines virtuelles).

Toutes les techniques présentées ici sont à des fins éducatives uniquement.

L'utilisation de ces outils sur des réseaux sans autorisation explicite est illégale.

Environnement requis

Machine	Rôle	OS recommandé
Attaquant	Lancer les outils d'audit	Kali linux
victimes	subir les scans et les attaques	Metasploitable ou windows
Routeur/GW	Passerelle simulée	Inclus dans le lan virtuel

Configuration réseau

mode réseau: host only ou en bridges

IP:

- 192.168.1.10 pour kali
- 192.168.1.20 pour Metasploit
- 192.168.1.3à pour Windows

NB: Toujours configurer le réseau en mode statique pour une stabilité, précision .

TÂCHE 1: Netdiscover

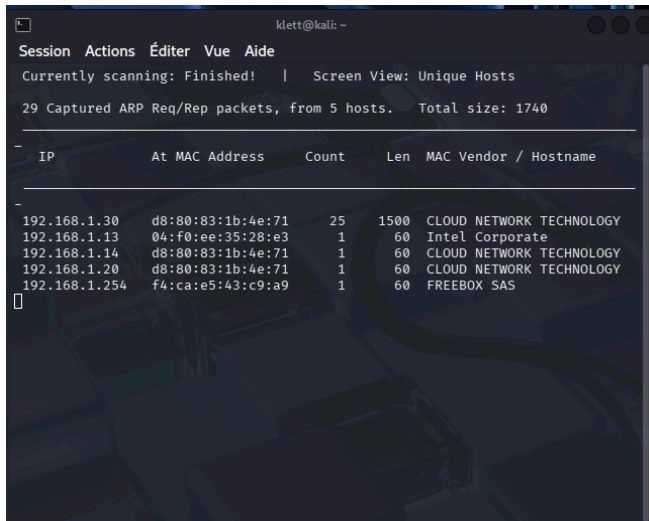
But: Découvrir les hôtes actifs sur un réseau en mode visible ou furtif.

Qu'est que Netdiscover?

Netdiscover est scanner ARP passif/actif conçu pour les réseaux sans fil et commutés. Il permet une détection des hôtes sans générer le trafic IP bruyant

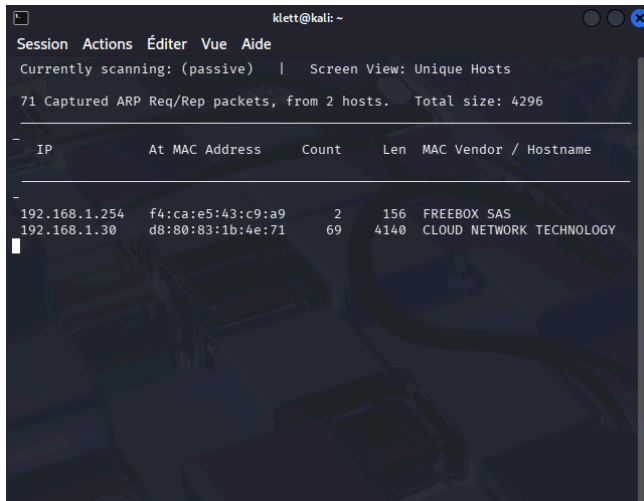
Mode actif (visible):

`sudo netdiscover -r 192.168.1.0/24`



Mode passif (Furtif):

`sudo netdiscover -p`



TÂCHES 2: Arping + Ettercap (ARP Spoofing)

But: Usurper l'identité du routeur via une attaque Man-in-the-Middle(ARP Poisoning).

Ici on fera croire à la victime que l'adresse MAC de l'attaquant est celle du routeur. Tout le trafic passera donc par la machine attaquante.

- a. **Activer le forwarding IP** pour que la connexion ne soit pas coupée. Cela transforme linux en **routeur, machine pivot ou relais réseau**.

```
echo 1 | sudo tee /proc/sys/net/ipv4/ip_forward
```

Pour vérifier:

```
cat /proc/sys/net/ipv4/ip_forward
```

résultat: 1

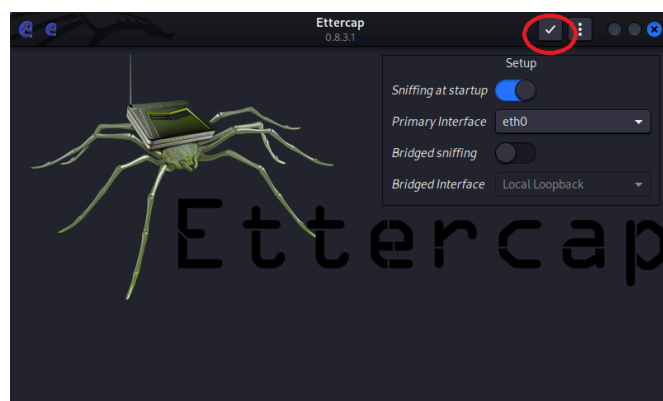


```
(klett@kali)-[~]  
$ cat /proc/sys/net/ipv4/ip_forward  
1
```

- b. **Lancer Ettercap en mode graphique** avec la commande **ettercap -G** ou dans application.

- c. Dans Ettercap:

- Sniff > Unified Sniffing → choisir l'interface réseau (ex: eth0) et lancez-la en appuyant sur le bouton que j'ai entouré en rouge ci-dessous



- Une fois lancée, cliquez sur le bouton ci-dessous et suivez les instructions se trouvant après l'image. Hosts > Scan for hosts



Hosts > Scan for hosts

Hosts > Hosts list → noter les IPs détectées

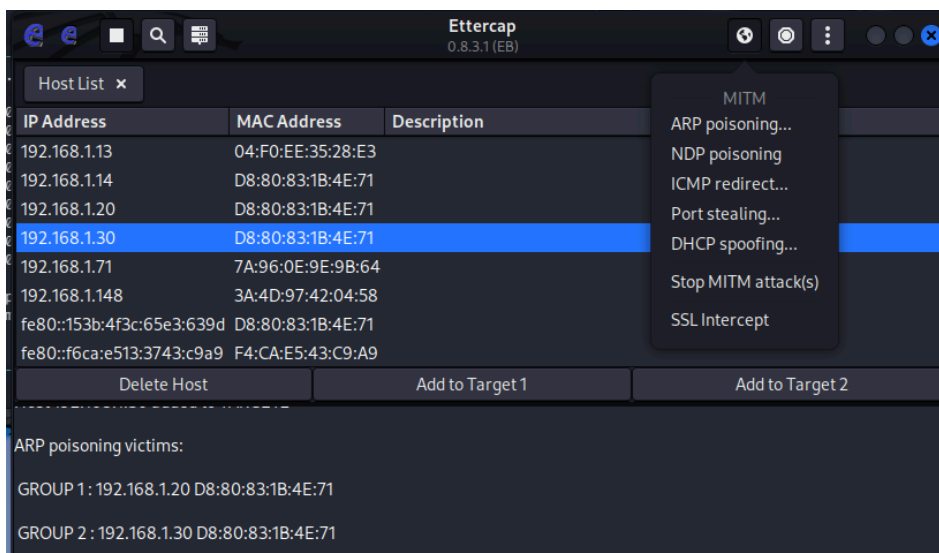
On obtiendra:

Host List ✕		
IP Address	MAC Address	Description
192.168.1.13	04:F0:EE:35:28:E3	
192.168.1.14	D8:80:83:1B:4E:71	
192.168.1.20	D8:80:83:1B:4E:71	
192.168.1.30	D8:80:83:1B:4E:71	
192.168.1.148	3A:4D:97:42:04:58	
192.168.1.194	76:87:64:DF:64:F5	
fe80::f6ca:e513:3743:c9a9	F4:CA:E5:43:C9:A9	
192.168.1.254	F4:CA:E5:43:C9:A9	
Delete Host Add to Target 1 Add to Target 2		

d. Définir les cibles

- IP du routeur : Add to target 1 = Metasploit
- IP de la victime: Add to target 2 = Windows

e. Lancer le ARP poisoning



f. Vérification du cache de la victime

Sur windows nous tapons

arp -a

```
C:\Users\voubou>arp -a

Interface : 192.168.1.30 --- 0xe
Adresse Internet    Adresse physique    Type
192.168.1.1         04-f0-ee-35-28-e3   dynamique
192.168.1.10        04-f0-ee-35-28-e3   dynamique
192.168.1.13        04-f0-ee-35-28-e3   dynamique
192.168.1.14        04-f0-ee-35-28-e3   dynamique
192.168.1.20        04-f0-ee-35-28-e3   dynamique
192.168.1.71        04-f0-ee-35-28-e3   dynamique
192.168.1.148       04-f0-ee-35-28-e3   dynamique
192.168.1.194       04-f0-ee-35-28-e3   dynamique
192.168.1.254       f4-ca-e5-43-c9-a9   dynamique
192.168.1.255       ff-ff-ff-ff-ff-ff   statique
224.0.0.22          01-00-5e-00-00-16   statique
224.0.0.251         01-00-5e-00-00-fb   statique
224.0.0.252         01-00-5e-00-00-fc   statique
239.255.255.250     01-00-5e-7f-ff-fa   statique
```

Nous constatons ici que l'adresse mac est identique sur les appareils se trouvant dans le même réseau.

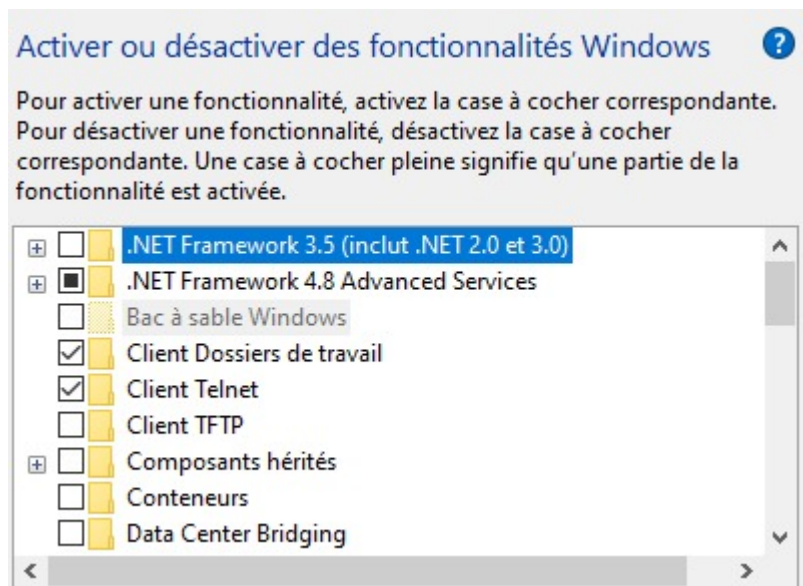
TÂCHES 3: Wireshark + Telnet

But: Démontrer que telnet transmet les en clair

a. Activer le service telnet

Sur Windows 10, Telnet n'est pas activé par défaut mais il est bien présent dans le système. Voici comment le trouver et l'activer :

1. Ouvre le Panneau de configuration → Programmes → Programmes et fonctionnalités.
2. Cliquez sur Activer ou désactiver des fonctionnalités Windows (à gauche).
3. Dans la liste, cochez Client Telnet.



4. Cliquez sur **OK**, puis laissez Windows installer le composant.
5. Dans le cmd taper telnet

Après installation de telnet sur notre windows nous n'arrivons à voir le port 23

```
C:\Users\voubou>netstat -an | find "23"
UDP    0.0.0.0:123          *:*
UDP    [::]:123            *:*
```

Lors de la vérification dans service on remarque que nous n'avons pas de server telnet donc pour la suite on utilisera metasploit 2 comme serveur telnet.

Pensez à vérifier si le port telnet est visible avec : `sudo netstat -tulpn | grep 23`

b. Lancer Wireshark sur la machine de l'attaquant et démarrer une capture sur l'interface réseau .

c. Depuis kali se connecter en telnet:

telnet 192.168.1.20

d. Analyser la capture:

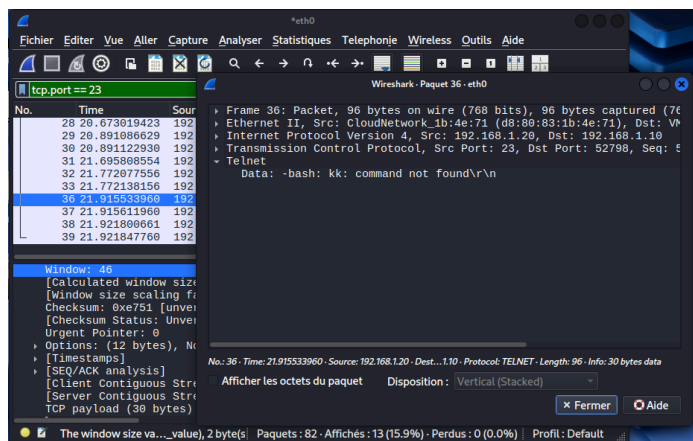
Avec le filtre **telnet** ou **tcp.port == 23** on constate que ce que nous écrivons est en clair.

```
6
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ kk
-bash: kk: command not found
```

Résultat sur Wireshark



TÂCHE 4: NMAP

But : Scanner les ports, détecter les services et identifier l'OS d'une cible.

a. Vérifier/installer nmap:

nmap --version

sudo apt install nmap s'il n'existe pas

b. Lister les ports tcp actifs:

nmap -sT <IP_cible>

```
(klett@kali) ~$ nmap -sT 192.168.1.30
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-13 17:59 CEST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.1.30
Host is up (0.0079s latency).
Not shown: 997 closed tcp ports (conn-refused)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
MAC Address: D8:80:83:1B:4E:71 (Cloud Network Technology Singapore PTE.)

Nmap done: 1 IP address (1 host up) scanned in 0.91 seconds
```

c. Surveiller un port tcp spécifique:

nmap -p 445 <IP_cible>

```
(klett@kali)-[~]
$ nmap -p 445 192.168.1.30
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-13 18:08 CEST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.1.30
Host is up (0.20s latency).

PORT      STATE SERVICE
445/tcp   open  microsoft-ds
MAC Address: D8:80:83:1B:4E:71 (Cloud Network Technology Singapore PTE.)

Nmap done: 1 IP address (1 host up) scanned in 0.69 seconds
```

d. Surveiller un port UDP spécifique:

- On va d'abord scanner les ports ouverts:

nmap -sU <IP_cible>

```
Nmap scan report for 192.168.1.30
Host is up (0.045s latency).
Not shown: 991 closed udp ports (port-unreach)
PORT      STATE SERVICE
123/udp   open|filtered ntp
137/udp   open          netbios-ns
138/udp   open|filtered netbios-dgm
500/udp   open|filtered isakmp
1900/udp  open|filtered upnp
4500/udp  open|filtered nat-t-ike
5050/udp  open|filtered mmcc
5353/udp  open|filtered zeroconf
5355/udp  open|filtered llmnr
MAC Address: D8:80:83:1B:4E:71 (Cloud Network Technology Singapore PTE.)

Nmap done: 1 IP address (1 host up) scanned in 925.98 seconds
```

- Ensuite sélectionné un port spécifique:

nmap -sU -p 123 <IP_cible>

```
(klett@kali)-[~]
$ nmap -sU -p 123 192.168.1.30
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-13 18:30 CEST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.1.30
Host is up (0.19s latency).

PORT      STATE SERVICE
123/udp   open|filtered ntp
MAC Address: D8:80:83:1B:4E:71 (Cloud Network Technology Singapore PTE.)

Nmap done: 1 IP address (1 host up) scanned in 2.67 seconds
```

- Scanner une plage de port:

nmap -p 20-100 <IP_cible>

e.

1. TCP stealth scan :

sudo nmap -sS <IP_cible>

```
$ sudo nmap -sS 192.168.1.30
[sudo] Mot de passe de klett :
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-15 17:25 CEST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.1.30
Host is up (0.032s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
MAC Address: D8:80:83:1B:4E:71 (Cloud Network Technology Singapore PTE.)

Nmap done: 1 IP address (1 host up) scanned in 1.98 seconds
```

2. ACK Scan (détection de firewall):

- **sudo nmap -sA 192.168.1.30**

- Observer le trafic **TCP dump**:

sudo tcpdump -i eth0 'tcp and src host 192.168.1.10' -n

NB: Le scan SYN Stealth (-sS) est le type de scan le plus utilisé. Il est rapide et n'établit pas complètement la connexion TCP, ce qui lui permet parfois d'éviter certains journaux applicatifs.

Cependant, les IDS/IPS modernes détectent facilement ce type de scan grâce aux modèles de connexions semi-ouvertes (half-open connections).

f. Détection du système d'exploitation et des versions de services:

sudo nmap -A 192.168.1.

```
(klett@kali)~$ sudo nmap -A 192.168.1.30
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-15 17:40 CEST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.1.30
Host is up (0.0058s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc      Microsoft Windows RPC
139/tcp    open  netbios-ssn Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
MAC Address: D8:80:83:1B:4E:71 (Cloud Network Technology Singapore PTE.)
Device type: general purpose
Running: Microsoft Windows 10
OS CPE: cpe:/o:microsoft:windows_10
OS details: Microsoft Windows 10 1709 - 21H2
Network Distance: 1 hop
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_ smb2-time:
|   date: 2026-05-15T15:49:31
|   start dates: N/A
|_ nbstat: NetBIOS name: DESKTOP-14LKR05, NetBIOS user: <unknown>, NetBIOS MAC
|   00:0C:29:7F:C3:19 (VMware)
|_ smb2-security-mode:
|   311:
|     Message signing enabled but not required
|_ clock-skew: -4s

TRACEROUTE
HOP RTT      ADDRESS
0  5.80 ms  192.168.1.30

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 16.47 seconds
```

g. Découverte d'hôtes:

1. Scan d'un réseau entier:

nmap -sn 192.168.1.0/24

sn: scan ping uniquement (pas de scan de ports)

```
(klett@kali)~$ nmap -sn 192.168.1.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-15 17:55 CEST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --
Nmap scan report for 192.168.1.13
Host is up (0.18s latency).
MAC Address: D8:80:83:1B:4E:71 (Cloud Network Technology Singapore PTE.)
Nmap scan report for 192.168.1.14
Host is up (0.0012s latency).
MAC Address: 04:F0:EE:35:28:E3 (Unknown)
Nmap scan report for 192.168.1.30
Host is up (0.26s latency).
MAC Address: D8:80:83:1B:4E:71 (Cloud Network Technology Singapore PTE.)
Nmap scan report for 192.168.1.92
Host is up (0.24s latency).
MAC Address: 0A:00:2A:15:09:6E (Unknown)
Nmap scan report for 192.168.1.117
Host is up (0.22s latency).
MAC Address: 3C:0B:59:B4:C5:63 (Unknown)
Nmap scan report for 192.168.1.254
Host is up (0.0052s latency).
MAC Address: F4:CA:E5:43:C9:A9 (Freebox SAS)
Nmap scan report for 192.168.1.10
Host is up.
Nmap done: 256 IP addresses (7 hosts up) scanned in 4.59 seconds
```


2. Scan d'une plage d'adresse:

nmap 192.168.1.1-30

```
(klett@kali) ~  
$ nmap 192.168.1.1-30  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-15 18:05 CEST  
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using  
Nmap scan report for 192.168.1.13  
Host is up (0.100s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
3306/tcp  open  mysql  
6646/tcp  open  unknown  
MAC Address: D8:80:83:1B:4E:71 (Cloud Network Technology Singapore PTE.)  
  
Nmap scan report for 192.168.1.14  
Host is up (0.00036s latency).  
All 1000 scanned ports on 192.168.1.14 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
MAC Address: 04:F0:EE:35:28:E3 (Unknown)  
  
Nmap scan report for 192.168.1.20  
Host is up (0.0047s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  cproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc
```

```
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc  
8009/tcp  open  ajp13  
8180/tcp  open  unknown  
MAC Address: D8:80:83:1B:4E:71 (Cloud Network Technology Singapore PTE.)  
  
Nmap scan report for 192.168.1.30  
Host is up (0.055s latency).  
Not shown: 997 closed tcp ports (reset)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
MAC Address: D8:80:83:1B:4E:71 (Cloud Network Technology Singapore PTE.)  
  
Nmap scan report for 192.168.1.10  
Host is up (0.000094s latency).  
All 1000 scanned ports on 192.168.1.10 are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
  
Nmap done: 30 IP addresses (5 hosts up) scanned in 26.43 seconds
```

3. Scan depuis un fichier:

- nano target.txt

```
GNU nano 8.6  
192.168.1.1  
192.168.1.10  
192.168.1.20  
192.168.1.30  
Session #  
Contact: i  
Login with  
metasploit  
Password:  
Login into  
metasploit  
Password:  
Last login:  
Linux met.  
$
```

- Lancer le scan:

nmap -iL targets.txt

```
(klett@kali)~$ nmap -iL targets.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-15 18:15 CEST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --
Nmap scan report for 192.168.1.20
Host is up (0.0061s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8000/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: D8:80:83:1B:4E:71 (Cloud Network Technology Singapore PTE.)

Nmap scan report for 192.168.1.30
Host is up (0.066s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
MAC Address: D8:80:83:1B:4E:71 (Cloud Network Technology Singapore PTE.)
```

h. Détection des protocoles réseau supportés :

sudo nmap -sO 192.168.1.20

```
(klett@kali)~$ sudo nmap -sO 192.168.1.30
[sudo] Mot de passe de klett :
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-15 18:22 CEST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --
Nmap scan report for 192.168.1.30
Host is up (0.067s latency).
Not shown: 252 open/filtered n/a protocols (no-response)
PROTOCOL STATE SERVICE
1         open  icmp
6         open  tcp
17        open  udp
132       closed sctp
MAC Address: D8:80:83:1B:4E:71 (Cloud Network Technology Singapore PTE.)

Nmap done: 1 IP address (1 host up) scanned in 7.53 seconds
```

TÂCHES 5: Le ping de la mort

But: Saturer la cible avec des requêtes ICMP Echo et comparer les débits de paquets selon les modes de flood.

man ping pour voir le manuel de ping

Pour le ping de la mort il faut suivre la numérotation suivante.

1. Ping flood avec -f: Envoie de paquet invisible:

sudo ping -f 192.168.1.30

Après 5 seconde faire:

ctrl c

```
(klett@kali)-[~]
$ sudo ping -f 192.168.1.30

PING 192.168.1.30 (192.168.1.30) 56(84) bytes of data.
... ^C
  192.168.1.30 ping statistics ---
1662 packets transmitted, 1659 received, 0.180505% packet loss, time 9608ms
rtt min/avg/max/mdev = 2.883/5.765/168.070/9.197 ms, pipe 11, ipg/ewma 5.784/12.014 ms
```

2. Préchargement de paquet avec -l:

sudo ping -l 1662 -c 1662 192.168.1.30

```
ping: WARNING: probably, rcvbuf is not enough to hold preload
PING 192.168.1.30 (192.168.1.30) 56(84) bytes of data.
64 bytes from 192.168.1.30: icmp_seq=1 ttl=128 time=32.7 ms
64 bytes from 192.168.1.30: icmp_seq=2 ttl=128 time=32.5 ms
64 bytes from 192.168.1.30: icmp_seq=3 ttl=128 time=31.8 ms
64 bytes from 192.168.1.30: icmp_seq=4 ttl=128 time=31.7 ms
64 bytes from 192.168.1.30: icmp_seq=5 ttl=128 time=31.4 ms
64 bytes from 192.168.1.30: icmp_seq=6 ttl=128 time=31.3 ms
64 bytes from 192.168.1.30: icmp_seq=7 ttl=128 time=31.2 ms
64 bytes from 192.168.1.30: icmp_seq=8 ttl=128 time=31.1 ms
64 bytes from 192.168.1.30: icmp_seq=9 ttl=128 time=30.0 ms
64 bytes from 192.168.1.30: icmp_seq=10 ttl=128 time=29.9 ms
64 bytes from 192.168.1.30: icmp_seq=11 ttl=128 time=29.8 ms
64 bytes from 192.168.1.30: icmp_seq=12 ttl=128 time=29.7 ms
64 bytes from 192.168.1.30: icmp_seq=13 ttl=128 time=29.6 ms
64 bytes from 192.168.1.30: icmp_seq=14 ttl=128 time=29.2 ms
64 bytes from 192.168.1.30: icmp_seq=15 ttl=128 time=29.1 ms
64 bytes from 192.168.1.30: icmp_seq=16 ttl=128 time=29.1 ms
64 bytes from 192.168.1.30: icmp_seq=17 ttl=128 time=29.0 ms
64 bytes from 192.168.1.30: icmp_seq=18 ttl=128 time=28.8 ms
64 bytes from 192.168.1.30: icmp_seq=19 ttl=128 time=28.1 ms
64 bytes from 192.168.1.30: icmp_seq=20 ttl=128 time=28.0 ms
64 bytes from 192.168.1.30: icmp_seq=21 ttl=128 time=27.9 ms
64 bytes from 192.168.1.30: icmp_seq=22 ttl=128 time=27.8 ms
64 bytes from 192.168.1.30: icmp_seq=23 ttl=128 time=27.7 ms
64 bytes from 192.168.1.30: icmp_seq=24 ttl=128 time=27.0 ms
64 bytes from 192.168.1.30: icmp_seq=25 ttl=128 time=26.9 ms
64 bytes from 192.168.1.30: icmp_seq=26 ttl=128 time=26.7 ms
64 bytes from 192.168.1.30: icmp_seq=27 ttl=128 time=26.6 ms
64 bytes from 192.168.1.30: icmp_seq=28 ttl=128 time=26.5 ms
64 bytes from 192.168.1.30: icmp_seq=29 ttl=128 time=26.2 ms
64 bytes from 192.168.1.30: icmp_seq=30 ttl=128 time=26.1 ms
64 bytes from 192.168.1.30: icmp_seq=31 ttl=128 time=26.0 ms
64 bytes from 192.168.1.30: icmp_seq=41 ttl=128 time=21.8 ms
64 bytes from 192.168.1.30: icmp_seq=42 ttl=128 time=23.0 ms
64 bytes from 192.168.1.30: icmp_seq=43 ttl=128 time=22.9 ms
64 bytes from 192.168.1.30: icmp_seq=44 ttl=128 time=22.8 ms
64 bytes from 192.168.1.30: icmp_seq=45 ttl=128 time=22.8 ms
64 bytes from 192.168.1.30: icmp_seq=46 ttl=128 time=23.7 ms
64 bytes from 192.168.1.30: icmp_seq=47 ttl=128 time=22.0 ms
64 bytes from 192.168.1.30: icmp_seq=48 ttl=128 time=21.7 ms
64 bytes from 192.168.1.30: icmp_seq=49 ttl=128 time=21.2 ms
64 bytes from 192.168.1.30: icmp_seq=50 ttl=128 time=22.3 ms
```

```
64 bytes from 192.168.1.30: icmp_seq=440 ttl=128 time=8.44 ms
64 bytes from 192.168.1.30: icmp_seq=441 ttl=128 time=9.39 ms
64 bytes from 192.168.1.30: icmp_seq=442 ttl=128 time=9.45 ms
64 bytes from 192.168.1.30: icmp_seq=443 ttl=128 time=10.1 ms
64 bytes from 192.168.1.30: icmp_seq=444 ttl=128 time=10.0 ms
64 bytes from 192.168.1.30: icmp_seq=445 ttl=128 time=9.95 ms
64 bytes from 192.168.1.30: icmp_seq=446 ttl=128 time=9.67 ms
64 bytes from 192.168.1.30: icmp_seq=447 ttl=128 time=9.62 ms
64 bytes from 192.168.1.30: icmp_seq=448 ttl=128 time=9.59 ms
64 bytes from 192.168.1.30: icmp_seq=449 ttl=128 time=9.54 ms
64 bytes from 192.168.1.30: icmp_seq=450 ttl=128 time=9.48 ms
64 bytes from 192.168.1.30: icmp_seq=451 ttl=128 time=11.7 ms
64 bytes from 192.168.1.30: icmp_seq=452 ttl=128 time=12.3 ms
64 bytes from 192.168.1.30: icmp_seq=453 ttl=128 time=12.2 ms
64 bytes from 192.168.1.30: icmp_seq=1625 ttl=128 time=14.3 ms
64 bytes from 192.168.1.30: icmp_seq=1638 ttl=128 time=15.7 ms
64 bytes from 192.168.1.30: icmp_seq=1639 ttl=128 time=15.6 ms
64 bytes from 192.168.1.30: icmp_seq=1640 ttl=128 time=15.4 ms
64 bytes from 192.168.1.30: icmp_seq=1641 ttl=128 time=15.3 ms
64 bytes from 192.168.1.30: icmp_seq=1642 ttl=128 time=15.2 ms
64 bytes from 192.168.1.30: icmp_seq=1643 ttl=128 time=15.2 ms
64 bytes from 192.168.1.30: icmp_seq=1644 ttl=128 time=15.1 ms
64 bytes from 192.168.1.30: icmp_seq=1645 ttl=128 time=9.02 ms
64 bytes from 192.168.1.30: icmp_seq=1646 ttl=128 time=8.95 ms
64 bytes from 192.168.1.30: icmp_seq=1647 ttl=128 time=8.91 ms
64 bytes from 192.168.1.30: icmp_seq=1649 ttl=128 time=9.31 ms
64 bytes from 192.168.1.30: icmp_seq=1650 ttl=128 time=9.20 ms
64 bytes from 192.168.1.30: icmp_seq=1651 ttl=128 time=10.4 ms
64 bytes from 192.168.1.30: icmp_seq=1652 ttl=128 time=11.0 ms
64 bytes from 192.168.1.30: icmp_seq=1659 ttl=128 time=11.4 ms
64 bytes from 192.168.1.30: icmp_seq=1660 ttl=128 time=11.9 ms
64 bytes from 192.168.1.30: icmp_seq=1661 ttl=128 time=11.9 ms
64 bytes from 192.168.1.30: icmp_seq=1662 ttl=128 time=11.8 ms

— 192.168.1.30 ping statistics —
1662 packets transmitted, 463 received, 72.142% packet loss, time 651ms
rtt min/avg/max/mdev = 5.178/11.694/32.682/6.028 ms, pipe 1662
```

NB: Le véritable “Ping of Death” (CVE-1999-0128) utilisait des paquets ICMP supérieurs à 65 535 octets.

Une fois réassemblés, ces paquets provoquaient **un débordements de mémoire** (buffer overflow) et **un crash système**.

Les systèmes de nos jours bloquent désormais ces paquets fragmentés malformés.